Managing the Effects of an Unexpected Power Loss on NAND Flash Program and Erase Operations

Introduction

SLC NAND Flash memory is a widely adopted choice of storage media in embedded applications where it is used to store operating system images, software applications, user data, and system data. When a system is powered on, the code is normally transferred from NAND to system DRAM for execution.

If during use, an application requires the NAND Flash be programmed or erased, typically the Status Register can be read to verify that the Program or Erase operation completed successfully.
However, should an unexpected power loss event occur during a Program or Erase operation, not only is it possible that the operation did not complete, but the Status Register verification check may also have not occurred.

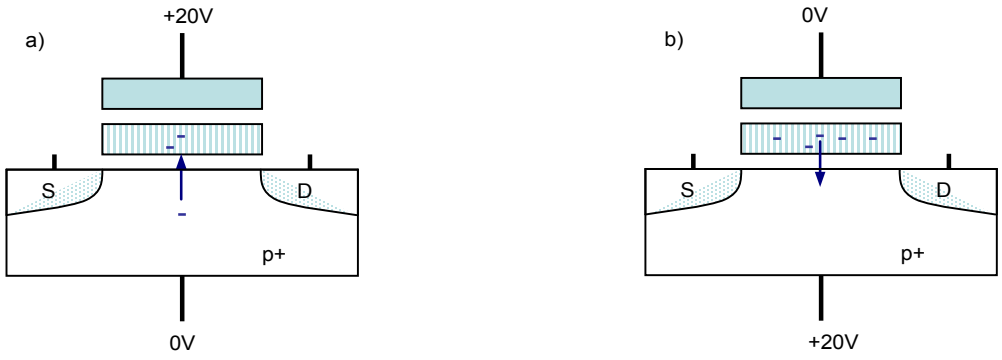
In some cases, if the removal of power is towards the completion of the Erase/Program operation, the flash management software may not find checksum errors after the power is back on. Since the Erase/Program operation is not completed with Status Register checks, the blocks which received the interrupted Program or Erase operations may have “unstable bit” behavior later on.

NAND Flash Program and Erase Internal Operations

Compared to reading or writing from/to a DRAM, altering the content of a NAND Flash memory is a slow process. Erasing a NAND Flash block typically takes 1ms. and programming a page typically takes 300us.

Structurally, each NAND Flash memory cell is a MOS transistor with a floating gate capable of retaining trapped charges. Electrons are injected into or extracted from the floating gate by mean of high electric field. The stored electrons in the floating gate have the effect of changing the cell’s threshold voltage V_{TH}. During a Program operation, the flash cells go through intermediate V_{TH} states until the necessary number of electrons has been stored into the floating gate. Conversely, during an erase operation, the cell’s threshold voltage V_{TH} goes through intermediate stages in the opposite direction.

Figure 1. Floating-gate FN-tunnel NAND (a) Flash Program and (b) Erase Mechanism

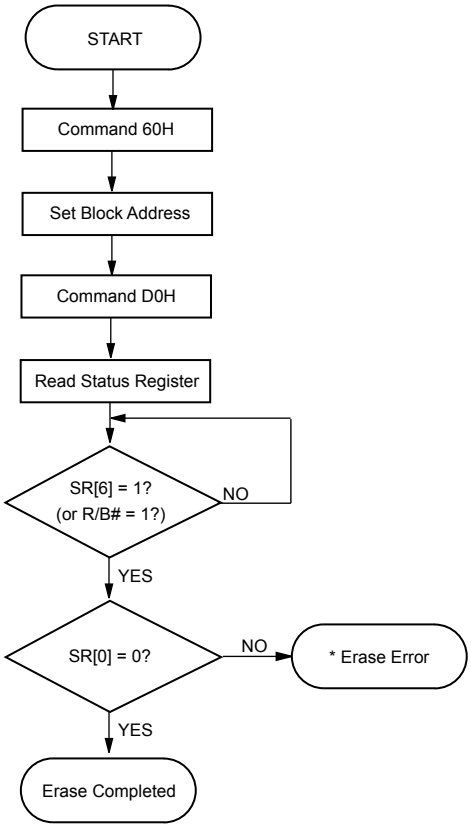


Note: The cell bias voltages shown in Figure 1 are typical and are for example purposes only.

Page Program and Block Erase are performed by sending specific commands to the NAND device. Once the command is issued, the memory becomes busy until the command is completed. When the Program/Erase operation is finished, the *"fail"* bit of the Status Register can be read (*"Figure 2. Erase Command: Algorithm Flow Chart"*) to verify that the Program or Erase operation completed successfully (please confirm this suggested method with your relevant NAND datasheet). If the success of the operation can't be verified, the blocks/pages should not be deemed as valid, even if the data seems to have been correctly programmed, or the block seem to have been correctly erased (i.e. read out as all "1").

Similarly, in case of an unexpected power loss, if a Program or Erase operation is interrupted, it is not always possible to check if the target block has been programmed or erased simply by reading it back after the power is turned back on. If the power interrupt occurs towards the very end of the operation, the block may seem to read out correctly, but some cells may not have reached the intended V_{TH} state which is designed to have sufficient margin for operation under all operating conditions. These affected bits may show eventually show signs of instability and may intermittently read incorrectly after multiple uses.

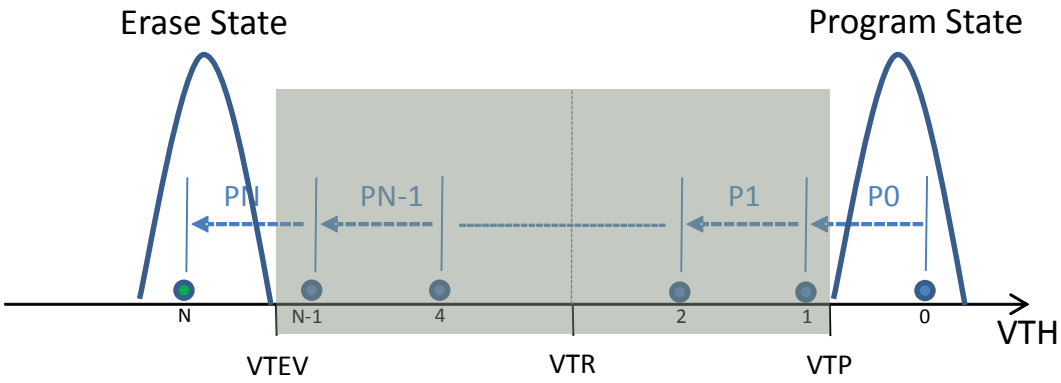
Figure 2. Erase Command: Algorithm Flow Chart



The internal process of erasing a NAND flash cell is illustrated as in *"Figure 2. Erase Command: Algorithm Flow Chart"* and explained as follows: Erasing a flash cell (already programmed to "0") is achieved by successively applying several electrical erase pulses to move the threshold voltage of the programmed cell from the state of high V_{TH} to the state of low V_{TH} as shown in *"Figure 3. Process of Erasing a Flash Cell"*.

In reality, all the cells within a particular block are subject to the same erase pulses and the resulting V_{TH} distributions of all the cells have a Gaussian shape as shown in *"Figure 3. Process of Erasing a Flash Cell"*. After each erase pulse, the V_T of any single cell is shifted to a lower threshold value and its state is checked though an Erase-Verify operation (which is similar to a Read operation) and compared against the *Erase-Verify level* V_{TEV}. An Erase is considered completed when, after several pulses, the V_T of the cell is below V_{TEV}. The *Erase-Verify level* V_{TEV} is designed to be sufficiently low, in order to guarantee that all the cells are read out as '1' (erased) in all temperature, voltage, and noise conditions.

Figure 3. Process of Erasing a Flash Cell



If an Erase operation is interrupted just before its completion (for example after or during PN-1 in *"Figure 3. Process of Erasing a Flash Cell"*), it may just so happen that the cell V_T is between V_{TEV} and the read level V_{TR}. Therefore, the cell may read as ''1' since its V_T is lower than the reading value V_{TR}. However, since there is not sufficient noise margin, random electrical noise may cause an erroneous reading. Affected cells may read as stable '1' initially, however, they are susceptible to failure after certain time when their V_T shifts toward the higher value under to normal operating conditions.

A similar situation can happen during the Programming process. It is the reverse process of bringing cell's V_{TH} from erased low-V_{TH} value to the programmed high-V_{TH} value, which corresponds to a programmed state.

Program and Erase Interrupted by a Power Loss Event

Sometimes, in the event of an unexpected power loss, it is not possible to determine whether an interrupted Erase/Program operation has been completed successfully by checking the Status Register. In such circumstances, affected blocks should be deemed as invalid, even if the data seems to read out correctly. In fact, as explained in the previous section, some flash cells may not have sufficient margin to guarantee a stable read-out for an extended period of time.

At the system level, Flash memory components are usually managed through low level software layers that deal with their unique characteristics. The MTD layer in Linux, for example, provides an abstraction layer to take care of specific hardware. UBI (Unsorted Block Images) is another software layer choice that can be added to take care of the block erase structure of NAND and other specific functions to NAND media, such as wear-leveling. File Systems generally sits on top of MTD or UBI.

To determine the importance of the described failure in application, it is necessary to consider:

- The probability of having a power loss during a Block Erase or Page Program in the application.
- The presence of a power loss detector in the system to prevent any flash Erase or Page Program operation should the system power falls below a certain threshold.
- The availability of a filter capacitance on the VCC power input that may extend the stability of the input power, thereby facilitating the completion of an Erase or Program operation in the event of an unexpected power loss.
- Macronix Memories are implemented with an internal power detector that disables any Program and Erase operation when the input power falls below a critical level. If the power down slope is long enough, this should help alleviate or even prevent the issue of interrupted Program and Erase operations as discussed in the previous sections.

In addition to the measures suggested above, it may also be possible to implement a system software solution to invalidate affected blocks which have not completed Erase/Program operations. In such cases, a new Erase/Program operation must be re-issued on the concerned blocks to restore them to a valid condition.

Summary

If a power loss condition happens during a NAND Program or Erase operation, and interrupts the operations before the Status Register can be verified, the affected Page or Block of memory may have partially programmed bits, which may exhibit unstable behavior during future Reads. Therefore, it might be necessary to review the low level software for proper management of the power loss conditions, to ensure that suspect pages and blocks can be identified and restored.

Revision History

Revision No.	Description	Page	Date
REV. 1	Initial Release	ALL	21st, Oct., 2014

Except for customized products which have been expressly identified in the applicable agreement, Macronix's products are designed, developed, and/or manufactured for ordinary business, industrial, personal, and/or household applications only, and not for use in any applications which may, directly or indirectly, cause death, personal injury, or severe property damages. In the event Macronix products are used in contradicted to their target usage above, the buyer shall take any and all actions to ensure said Macronix's product qualified for its actual use in accordance with the applicable laws and regulations; and Macronix as well as it's suppliers and/or distributors shall be released from any and all liability arisen therefrom.

Copyright© Macronix International Co., Ltd. 2014. All rights reserved, including the trademarks and tradename thereof, such as Macronix, MXIC, MXIC Logo, MX Logo, Integrated Solutions Provider, NBit, Nbit, NBiit, Macronix NBit, eLite-Flash, HybridNVM, HybridFlash, XtraROM, Phines, KH Logo, BE-SONOS, KSMC, Kingtech, MXSMIO, Macronix vEE, Macronix MAP, Rich Audio, Rich Book, Rich TV, and FitCAM. The names and brands of third party referred thereto (if any) are for identification purposes only.

For the contact and order information, please visit Macronix's Web site at: http://www.macronix.com

MACRONIX INTERNATIONAL CO., LTD. reserves the right to change product and specifications without notice.